velocities of the molecules that have the same, overall, gross features and appearance. If the molecules' configuration were changed from one point in a given region of phase space to another point in the same region, a macroscopic assessment would find the two configurations indistinguishable. Now, rather than counting the number of points in a given region—the most direct analog of counting the number of different page rearrangements, but something that will surely result in an infinite answer—physicists define entropy in terms of the *volume* of each region in phase space. A larger volume means more points and hence higher entropy. And a region's volume, even a region in a higher-dimensional space, is something that can be given a rigorous mathematical definition. (Mathematically, it requires choosing something called a measure, and for the mathematically inclined reader, I'll note that we usually choose the measure which is uniform over all microstates compatible with a given macrostate—that is, each microscopic configuration associated with a given set of macroscopic properties is assumed to be equally probable.)

10. Specifically, we know one way in which this *could* happen: if a few days earlier the CO_2 was initially in the bottle, then we know from our discussion above that if, right now, you were to simultaneously reverse the velocity of each and every CO_2 molecule, and that of every molecule and atom that has in any way interacted with the CO_2 molecules, and wait the same few days, the molecules *would* all group back together in the bottle. But this velocity reversal isn't something that can be accomplished in practice, let alone something that is likely to happen of its own accord. I might note, though, that one can prove mathematically that if you wait long enough, the CO_2 molecules will, of their own accord, *all find their way back into the bottle.* A result proven in the 1800s by the French mathematician Joseph Liouville can be used to establish what is known as the Poincaré recurrence theorem. This theorem shows that, if you wait long enough, a system with a finite energy and confined to a finite spatial volume (like CO_2 molecules in a closed room) will return to a state arbitrarily close to its initial state (in this case, CO_2 molecules all situated in the Coke bottle). The catch is how long you'd have to wait for this to happen. For systems with all but a small number of constituents, the theorem shows you'd typically have